

Joseph Barbosa

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EDUCATION

University of Waterloo (Double Degree with Wilfrid Laurier University) GPA: 3.98 / 4.0 (92%)
Bachelor of Computer Science (AI Specialization, Stats Minor) & Bachelor of Business Administration Sep. 2025 – Present

EXPERIENCE

- Undergraduate Research Assistant** Incoming Sep. 2026
University of Waterloo Waterloo, ON
- Evaluating ASR models on a **380-hour** code-switched Kazakh-Russian medical transcript corpus under Professor Helen Chen; analyzing transcription errors under language switching, improving robustness of AI-assisted clinical documentation models.
- Software Engineering Intern** May 2026 – Aug. 2026
Scotiabank Toronto, ON
- Built Flowpilot: a VLM-powered ReAct QA agent using Set-of-Marks grounding for live UI test automation. Achieved a **78%** one-shot pass rate on a held-out **33-case** evaluation suite. Reduced QA time from **2h to 20m** across **5 services**.
 - Designed AI architecture for ProfileAI (shipping FY27, **50K clients/day**): vector search, pre-filtering, and prompt caching pipelines; collaborated with internal and MongoDB engineers to design production-scale inference and retrieval strategies.
- Case Team Delegate** Jun. 2026 – Present
Wilfrid Laurier University Waterloo, ON
- Selected as a delegate for Laurier's digital strategy and international case teams; competing in 5–12h practices twice weekly.
 - Designed an AI-assisted team workflow (mockups, transcriptions, and solutioning), adopted by international teammates.
- Software Engineering Intern** Jan. 2026 – Apr. 2026
ESGTree Waterloo, ON
- Built a production NL-to-SQL agent over a **98-table** MySQL database using ReAct reasoning, schema retrieval, & join-aware prompting; achieved **90.9%** accuracy on **77** eval queries, crediting self-correction (**+13.0pp**) & RAG (**+7.8pp**) gains.
 - Engineered production backend with SQL validation, read-only DB access, and multi-tenant isolation via JWT; designed a 3-model cost/latency tier and Docker-sandboxed Python execution for data analytics through a Next.js interface.
- Software Development Intern** Jun. 2024 – Dec. 2024
FuturIQ Brampton, ON
- Built an end-to-end ML pipeline for housing price prediction: automated Selenium data ingestion of **1,000+** current listings and engineered **8** features, deploying the regularized polynomial model via Flask/React with realtime automated retraining.

PROJECTS

- Studeal** | *Python, Chrome DevTools Protocol, Celery, Redis, Postgres + pgvector, AWS Bedrock* | [GitHub](#)
- Built a monetized deal-hunting platform where autonomous LLM browser agents search **10** secondhand marketplaces from conversational specifications, aggregate findings into a shared corpus, and alert users through a Next.js dashboard.
 - Diagnosed context rot in a single-agent ReAct baseline and replaced it with context-isolated navigator/extractor agents using overlap-chunked extraction, yielding **2.7×** more unique listings, **63%** faster runtimes, and **8×** per-page recall (13 → 106).
 - Eliminated site-specific scraper code using Chrome DevTools Protocol accessibility-tree perception, enabling **5/8** previously unseen marketplaces to return valid listings on first exposure.
 - Built a cold-start-free recommendation pipeline combining conversational intent elicitation, pgvector retrieval, and similarity gated deduplication to surface newly covered listings from the shared corpus in seconds instead of minutes.
- Scholar** | *Python, FastAPI, Next.js, sentence-transformers, OpenAI* | [GitHub](#) tryscholar.com
- Built a **multi-agent** research assistant over **200M+** papers by decomposing user queries into subtopics researched by parallel agents leveraging recursive, bounded-depth planning. Deployed via MCP and a **Next.js** application to **300+** users.
 - Engineered a two-stage retrieve-and-rerank system (BGE-small bi-encoder → MiniLM cross-encoder), lifting **nDCG@5** by **10.8%** over single-stage dense (bi-encoder only) retrieval on an LLM-judged benchmark of **20** handwritten eval queries.
 - Hardened for production with hallucinated-citation stripping, retry/replanning, and graceful degradation on model failure.
- PantryPal** | *Python, PyTorch, QLoRA, DPO, XGBoost* | [GitHub](#) [Blog Post](#)
- Investigated whether weak domain classifiers could replace human preference labels for LLM alignment; preprocessed **10K** RecipeNLG examples to train **18** XGBoost models across dietary categories, generating **1,200+** DPO preference pairs.
 - Fine-tuned Llama 3.2 3B with QLoRA SFT on **8K** recipes, reaching **99.7% keyword compliance** vs. **56.8%** GPT-4o.

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL, MySQL, SQLite), TypeScript, JavaScript, HTML/CSS, C
AI / ML: PyTorch, Hugging Face Transformers, PEFT, LangGraph, LangChain, MCP, fine-tuning, RAG, AI evals, OpenAI API
Frameworks: React, Next.js, FastAPI, Flask, Node.js
Libraries: pandas, NumPy, scikit-learn, XGBoost, Pydantic, SQLAlchemy, Selenium, Playwright
Databases / Cloud: PostgreSQL, SQLite, Redis, pgvector, FAISS, Azure, AWS
Tools: Git, Docker, Bash, Claude Code, CI/CD, GitHub Actions
Awards: President's Gold Scholarship, Loblaw National Scholarship, Stanford ML Specialization, \$3,000+ in Case Winnings